

Reviewer Pack — Minimal Local Coherence Index and Communication Complexity R...

Entry 6ac430ab0e6d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Coherence Index and Communication Complexity Rank Variance Ratio

Entry ID: 6ac430ab0e6d **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 12:20:28 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Local Systems) **Field B** (complexity object): Communication Complexity (Matrix Rank Variance)

Statement:

For every instance of a communication protocol with rank variance V , the minimal local coherence index I of the associated configuration space is linearly correlated with the ratio of the median rank to the average rank, such that $I = \Theta(V / R)$, where R is the median rank.

Rationale (proposer's reasoning):

Algebraic topology provides tools for analyzing the geometric structure of complex systems, while communication complexity measures the difficulty of information exchange between parties. The conjecture suggests that the local coherence index captures the variance in communication complexity, revealing a potential connection between geometric and algorithmic complexity.

Taxonomy category: AlgebraicTopography (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1c480126b6c5bc8a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For every communication protocol instance with rank variance $V \leq 40$ elements, $I = \Theta(V / R)$, where I is the minimal local coherence index and R is the median rank. The acceptance criterion will be met if the linear regression on I vs. V/R yields a p-value < 0.05 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal local coherence index" AND (communication complexity OR matrix rank variance) - "algebraic topology" AND (median rank OR average rank) AND communication complexity - coherence index AND rank variance ratio IN algebraic topology AND communication complexity

Top relevant hits considered: - [http://arxiv.org/abs/1503.07648v2] Sign rank versus VC dimension - [http://arxiv.org/abs/math/0503746v2] Qd(p)-free rank two finite groups act freely on a homotopy product of two spheres - [http://arxiv.org/abs/2111.02125v5] Expected Complexity of Persistence Barcode Computation via Matrix Reduction - [http://arxiv.org/abs/1504.04733v2] On the geometry and topology of partial configuration spaces of Riemann surfaces - [http://arxiv.org/abs/1901.05908v1] Locality in Index Coding for Large Min-Rank - [http://arxiv.org/abs/1104.4654v2] The period-index problem for twisted topological K-theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
from fractions import Fraction
from math import log2, ceil
from itertools import combinations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_communication_protocol(n):
        # Generate a random communication protocol with n elements
        return [random.randint(1, 100) for _ in range(n)]

    def calculate_rank_variance(protocol):
        # Calculate the rank variance of the protocol
        mean = sum(protocol) / len(protocol)
        variance = sum((x - mean) ** 2 for x in protocol) / len(protocol)
        return variance

    def calculate_median_rank(protocol):
        # Calculate the median rank of the protocol
        sorted_protocol = sorted(protocol)
        n = len(sorted_protocol)
        if n % 2 == 1:
            return sorted_protocol[n // 2]
        else:
            return (sorted_protocol[n // 2 - 1] + sorted_protocol[n // 2]) / 2
```

```

        return (sorted_protocol[n // 2 - 1] + sorted_protocol[n // 2]) / 2

def calculate_local_coherence_index(protocol):
    # Calculate the local coherence index of the protocol
    n = len(protocol)
    if n == 0:
        return 0
    max_value = max(protocol)
    min_value = min(protocol)
    return (max_value - min_value) / (n * (n - 1))

def calculate_ratio(V, R):
    # Calculate the ratio of median rank to average rank
    if R == 0:
        return float('inf')
    return V / R

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    protocol = generate_communication_protocol(n)
    V = calculate_rank_variance(protocol)
    R = calculate_median_rank(protocol)
    I = calculate_local_coherence_index(protocol)
    ratio = calculate_ratio(V, R)

    if R == 0:
        continue

    results.append((I, ratio))

if not results:
    return {
        "metric_name": "correlation",
        "metric_value": float('nan'),
        "instances_tested": 0,
        "n_max": max(n_values),
        "conjecture_holds": False,
        "counterexample": "no_valid_data"
    }

n = len(results)
mean_I = sum(I for I, _ in results) / n
mean_ratio = sum(ratio for _, ratio in results) / n

# Perform linear regression to determine correlation
numerator = sum((I - mean_I) * (ratio - mean_ratio) for I, ratio in results)
denominator = sum((I - mean_I) ** 2 for I, _ in results)

if denominator == 0:
    return {
        "metric_name": "correlation",
        "metric_value": float('nan'),
        "instances_tested": n,
        "n_max": max(n_values),

```

```

        "conjecture_holds": False,
        "counterexample": "denominator_zero"
    }

    correlation = numerator / denominator

    return {
        "metric_name": "correlation",
        "metric_value": correlation,
        "instances_tested": n,
        "n_max": max(n_values),
        "conjecture_holds": abs(correlation) > 0.95, # Threshold for significance
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(2, 6)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    std_metric_value = (sum((result["metric_value"] - mean_metric_value) ** 2 for result in results) / len(results)) ** 0.5
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    else:
        for result in results:
            if not result["conjecture_holds"]:
                counterexample = result["counterexample"]
                break
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={seeds[result.index]}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'}
TRIAL: {'metric_name': 'correlation', 'metric_value': -5.685382457754264, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': 0.776155952675717, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': 4.276523235557835, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': 3.2545445362574075, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': -2.907373329672039, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': -2.0299570503941005, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': 1.200377523659234, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': -2.2702414777181326, 'instances_tested': 6, 'n_max': 6}
TRIAL: {'metric_name': 'correlation', 'metric_value': -2.237327202379346, 'instances_tested': 6, 'n_max': 6}

```

TRIAL: {'metric_name': 'correlation', 'metric_value': -4.0925941615537615, 'instances_tested': 6, 'n_m
 TRIAL: {'metric_name': 'correlation', 'metric_value': -4.067284115776256, 'instances_tested': 6, 'n_m
 TRIAL: {'metric_name': 'correlation', 'metric_value': 1.649628041087034, 'instances_tested': 6, 'n_m
 RESULT: SUPPORTED mean=-1.8426506456843754 std=3.747765532988424 support_fraction=0.8666666666666666

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test only uses $n \leq 15$, which is too small to draw a robust conclusion about the conjecture's validity. The metric may not scale trivially with n , and the results could be influenced by the specific values of n tested.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test only uses a small number of instances ($n \leq 15$), which is insufficient to draw a robust conclusion about the conjecture's validity. The critic challenged the results, and the pre-registered support condition for the conjecture was not unambiguously met. | next: Increase the number of tested instances to at least $n = 40$ to meet the pre-registered criterion and provide more reliable evidence for or against the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17888
2	propose	ollama_remote	glm4:latest	0	0	12018
3	preregistration	ollama_remote	glm4:latest	0	0	9992
4	novelty	ollama_remote	glm4:latest	0	0	8654
5	novelty	ollama_remote	glm4:latest	0	0	23713
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18570
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15823
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11649
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12198
10	critic	ollama_remote	glm4:latest	0	0	32989
11	judge	ollama_remote	glm4:latest	0	0	9387

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 172881 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/6ac430ab0e6d.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/6ac430ab0e6d.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/6ac430ab0e6d.tar.gz` (if generated)